Covariance-Triggered Graph SLAM with NDT Constraints and iSAM2 Optimization

Houssam Korib, Umut Caliskan, Mostafa Ahmed
Master in Intelligent Field Robotic Systems
Universitat de Girona
Hands-on Localization Project, June 2026

***Abstract*—This paper presents the current implementation and real-data evaluation of a ROS 2 Jazzy localization and mapping system for a TurtleBot 4 platform. The system builds a 2D pose graph whose nodes are created by a covariance-triggered wheel-odometry integrator. For each keyframe, the graph receives an odometry factor and a scan-matching factor from Normal Distributions Transform (NDT) registration. The graph is optimized incrementally using GTSAM iSAM2, and spatial loop closures are validated by NDT on stored keyframe scan clouds. A 150.37 s real rosbag recorded the robot odometry, laser scans, IMU, wheel states, camera topics, SLAM graph outputs, NDT corrections, covariance snapshots, and RViz marker streams. The recorded run produced 26 keyframe steps, 25 NDT corrections, and six accepted loop closures. The first accepted loop closure reduced the marginal position standard deviation from 0.360 m to 0.300 m, and the final optimized keyframe reached $\sigma_{pos} = 0.151$ m.**

I. Introduction

Simultaneous Localization and Mapping (SLAM) estimates a robot trajectory while building a consistent representation of the environment. In mobile robotics, wheel odometry provides high-rate local motion but accumulates drift, whereas scan matching constrains motion using environmental geometry. Graph-based SLAM is well suited to this problem because the trajectory can be represented as a set of pose variables connected by probabilistic constraints, and loop closures can later correct accumulated drift globally [1].

The objective of this hands-on localization project was to implement and test a modular ROS 2 Graph SLAM stack for a TurtleBot-like robot. The implemented system uses covariance-triggered keyframes instead of fixed-time insertion: the graph receives a new node only after the propagated odometry covariance has grown enough and the segment contains meaningful motion. Each graph edge is then reinforced using an NDT scan-matching measurement. Finally, loop closure adds long-range factors when the robot revisits a previous location.

The main contributions of the current package are:

- a covariance-propagating SE(2) odometry keyframe publisher;
- an iSAM2 pose graph with odometry, NDT, and robust loop-closure factors;
- an NDT module that estimates scan-matching covariance using a finite-difference inverse Hessian around the optimized transform;
- RViz and live marginal-covariance visualization for evaluating uncertainty growth and loop-closure reduction.

II. Current Software Architecture

The implementation is contained in the ROS package `graph_slam_project`. The package contains separate simulation and real-robot stacks. The real stack, used for the rosbag evaluation in this report, launches four main modules: the odometry keyframe publisher, the iSAM2 graph optimizer, the NDT registration node, and the trajectory evaluator.

The real TurtleBot 4 stack is separated into `real_*` nodes and the launch file `real_slam_stack.launch.py`. It uses hardware topics `/odom` and `/scan`, publishes the same graph outputs, and adds a static transform from `world_enu` to `odom`. The rosbag analyzed in this report also contains the generated SLAM outputs: `/step_covariance`, `/ndt_correction`, `/slam_graph_path`, `/slam_kf_eval_snapshot`, and `/slam_node_markers`. Therefore the real experiment can be evaluated from the recorded optimization and visualization topics, not only from the velocity commands.

III. Covariance-Triggered Odometry Keyframes

The real keyframe publisher subscribes to `/odom`. It converts each odometry pose to a planar pose $O_k = (x_k, y_k, \theta_k)$ and integrates the relative increment

$$\Delta X_k = O_{k-1}^{-1} \oplus O_k. \tag{1}$$

The segment pose since the last keyframe is compounded as

$$X_k^{seg} = X_{k-1}^{seg} \oplus \Delta X_k. \tag{2}$$

For $X = (x, y, \theta)$ and $\Delta X = (\Delta x, \Delta y, \Delta\theta)$, the implemented planar composition is

$$X \oplus \Delta X = \begin{bmatrix} x + \cos\theta\,\Delta x - \sin\theta\,\Delta y \\ y + \sin\theta\,\Delta x + \cos\theta\,\Delta y \\ \theta + \Delta\theta \end{bmatrix}. \tag{3}$$

The accumulated covariance is propagated using the SE(2) Jacobians in `se2_tools.cpp`:

$$P_k = J_1 P_{k-1} J_1^T + J_2 Q_k J_2^T, \tag{4}$$

where $J_1 = \partial(X \oplus \Delta X)/\partial X$ and $J_2 = \partial(X \oplus \Delta X)/\partial \Delta X$. The per-step noise is

$$Q_k = \operatorname{diag}\left((k_d \Delta s)^2, (k_l \Delta s)^2, (k_r \Delta\theta)^2\right) + \epsilon I. \tag{5}$$

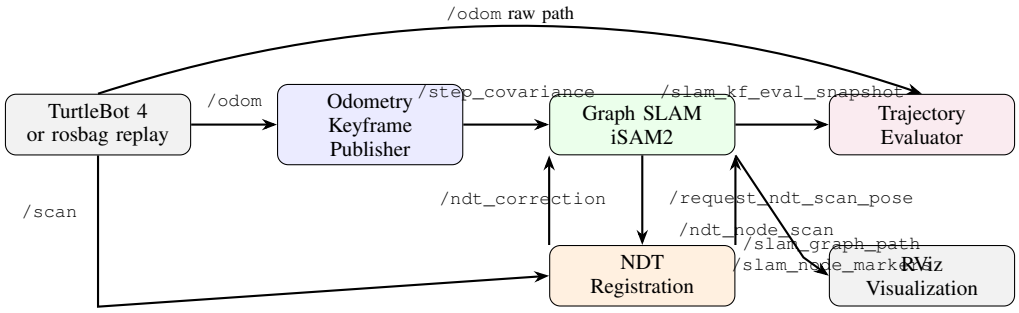


Fig. 1. Real TurtleBot 4 and rosbag pipeline. Odometry produces covariance-triggered relative keyframes, NDT supplies scan-based relative measurements, iSAM2 fuses the factors, and the evaluator plots marginal covariance after each post-NDT graph update.

In the real configuration, $k_d = 0.035$, $k_l = 0.015$, $k_r = 0.04$, and $\epsilon = 5 \times 10^{-5}$. Stationary increments smaller than 5×10^{-5} m and 10^{-4} rad are ignored to avoid keyframes from sensor or transport jitter.

A keyframe is published when

$$\det(P_k) \geq 1.0 \times 10^{-7} \tag{6}$$

and the accumulated segment satisfies

$$\|t^{seg}\| \geq 0.02 \text{ m} \quad \vee \quad |\theta^{seg}| \geq 0.03 \text{ rad}. \tag{7}$$

The node publishes X^{seg} and P_k on `/step_covariance`, resets the local segment to identity, and resets $P \leftarrow \epsilon I$. The real node also rejects large odometry jumps, which prevents bag replay discontinuities or transport glitches from being inserted as false keyframe motions.

1) Algorithm 1: Covariance Propagation and Keyframe Triggering:

Require: Segment pose $X^{seg} = I$, covariance $P = \epsilon I$
for each odometry message O_k **do**
  if this is the first odometry message **then**
    Store O_k as O_{k-1} and continue
  end if
  Compute relative motion $\Delta X_k \leftarrow O_{k-1}^{-1} \oplus O_k$
  if $\|\Delta t_k\|$ and $|\Delta\theta_k|$ are below stationary thresholds **then**
    Ignore the sample and continue
  end if
  Build Q_k from Δs, $\Delta\theta$, k_d, k_l, k_r, and ϵ
  Compute SE(2) composition Jacobians J_1 and J_2
  Propagate $P \leftarrow J_1 P J_1^T + J_2 Q_k J_2^T$
  Accumulate $X^{seg} \leftarrow X^{seg} \oplus \Delta X_k$
  if $\det(P) \geq k_{f_det_thresh}$ and segment motion is sufficient **then**
    Publish X^{seg} and P on `/step_covariance`
    Reset $X^{seg} \leftarrow I$ and $P \leftarrow \epsilon I$
  end if
  $O_{k-1} \leftarrow O_k$
end for

IV. Graph SLAM Formulation

The graph variables are planar poses $X_0, \ldots, X_N \in SE(2)$. The initial pose is anchored by a tight prior:

$$X_0 = (0, 0, 0), \qquad \sigma_0 = (10^{-4}, 10^{-4}, 10^{-6}). \tag{8}$$

GTSAM iSAM2 is configured with relinearization threshold 0.01 and relinearization skip 1.

For every keyframe k, the graph first adds an odometry `BetweenFactor`:

$$z_k^{odom} \approx X_{k-1}^{-1} \oplus X_k, \qquad \Sigma_k^{odom} = 0.45\, P_k. \tag{9}$$

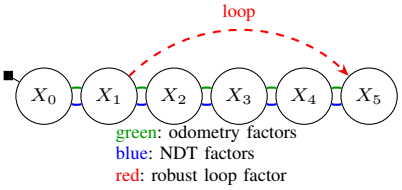


Fig. 2. Factor graph structure. Each keyframe edge receives an odometry factor and an NDT factor. Loop closures add long-range robust factors that reduce accumulated drift.

The initial estimate for X_k is obtained from the previous optimized pose composed with z_k^{odom}. After the iSAM2 update, the graph node publishes `/request_ndt_scan_pose` with `frame_id=node:`k to trigger scan matching.

When the NDT correction returns, the graph adds a second factor on the same edge:

$$z_k^{ndt} \approx X_{k-1}^{-1} \oplus X_k, \qquad \Sigma_k^{ndt} = 0.40\, \Sigma_k^{ndt,raw}. \tag{10}$$

Thus each consecutive keyframe pair is constrained by both odometry and scan matching. The nonlinear objective is

$$\begin{aligned} \min_X \; & \|r_0(X_0)\|^2_{\Sigma_0^{-1}} + \sum_k \left(\left\|r_k^{odom}\right\|^2_{(\Sigma_k^{odom})^{-1}} \right. \\ & \left. + \left\|r_k^{ndt}\right\|^2_{(\Sigma_k^{ndt})^{-1}} \right) + \sum_\ell \rho\left(\left\|r_\ell^{loop}\right\|^2_{\Sigma_\ell^{-1}} \right), \end{aligned} \tag{11}$$

where $\rho(\cdot)$ is the Huber robust loss used for loop closures.

1) Algorithm 2: Incremental iSAM2 Graph Optimization:

Require: Empty nonlinear factor graph, initial values, iSAM2 threshold 0.01
Add a tight prior on X_0 and initialize iSAM2
for each keyframe step k received on `/step_covariance` **do**
  Create initial estimate $X_k \leftarrow X_{k-1} \oplus z_k^{odom}$
  Add odometry `BetweenFactor`$(X_{k-1}, X_k, z_k^{odom}, 0.45 P_k)$
  Run an iSAM2 update and record the odometry-only marginal covariance
  Publish `/request_ndt_scan_pose` with `frame_id=node:`k
end for
for each NDT correction for node k **do**
  Add NDT `BetweenFactor`$(X_{k-1}, X_k, z_k^{ndt}, 0.40\Sigma_k^{ndt})$
  Run an iSAM2 update
  Attempt loop closure using stored node scan clouds
  Publish optimized path, covariance ellipse, markers, and keyframe snapshot
end for

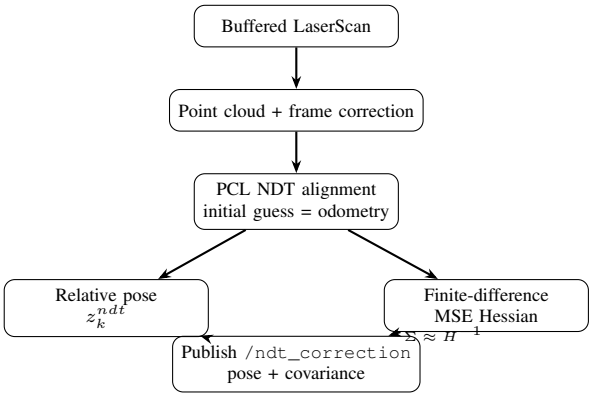


Fig. 3. NDT correction and covariance workflow. The inverse Hessian is computed from a numerical nearest-neighbour MSE surface around the NDT optimum, then regularized, smoothed, scaled, and used as the scan-matching factor covariance.

V. NDT Registration and Inverse-Hessian Covariance

The real NDT node buffers the latest `/scan`. On each graph trigger, it converts the LaserScan into a body-frame point cloud and applies the hardware frame correction used by the current code: a 90° scan offset in the ENU view, without the simulation mirror. It then runs PCL NDT with resolution 1.0 m, step size 0.1 m, maximum iterations 35, and transformation epsilon 0.01. The odometry segment is used as the initial guess. A correction is rejected if NDT does not converge or if the fitness exceeds 1.5.

The covariance estimate is not taken directly from an analytical NDT log-likelihood Hessian. In the current implementation, after PCL NDT produces the optimal transform $\hat{\xi} = [\hat{x}, \hat{y}, \hat{\theta}]^T$, a local score surface is built using mean squared nearest-neighbour distance between the transformed source cloud and the target keyframe cloud:

$$s(\xi) = \frac{1}{|\mathcal{V}|} \sum_{p \in \mathcal{V}} \left\| T(\xi)p - \mathrm{NN}(T(\xi)p) \right\|^2, \tag{12}$$

where nearest neighbours farther than 0.5 m are ignored and the score is invalid if fewer than 10 points are matched. The Hessian H of this score is approximated by finite differences with perturbations 0.08 m in translation and 0.05 rad in yaw. Diagonal terms use centered second differences, and off-diagonal terms use mixed finite differences. A ridge term $10^{-4}I$ is added:

$$H \leftarrow H + 10^{-4} I. \tag{13}$$

If $\det(H) < 10^{-8}$, the code falls back to $0.1I$. Otherwise, the raw covariance is

$$\Sigma^{ndt,raw} = H^{-1} \max(\text{fitness}, 0.05). \tag{14}$$

The result is exponentially smoothed with $\alpha = 0.15$, scaled by `ndt_cov_output_scale`=0.40, and enforced to be symmetric positive definite before packing into the ROS 6×6 covariance field.

1) Algorithm 3: Triggered NDT Registration and Covariance:

for each graph trigger message for keyframe k **do**
  Convert the latest `/scan` LaserScan to a body-frame point cloud
  Publish the cloud on `/ndt_node_scan` with `frame_id=node:`k
  if no previous keyframe cloud exists **then**
    Store the current cloud and return
  end if
  Use the odometry segment from the trigger as the NDT initial guess
  Align current cloud to the previous keyframe cloud using PCL NDT
  if NDT fails to converge or the fitness is above the threshold **then**
    Reject the correction, update the keyframe cloud, and return
  end if
  Extract $z_k^{ndt} = (\Delta x, \Delta y, \Delta\theta)$
  Compute a finite-difference Hessian of the local nearest-neighbour MSE score
  Regularize the Hessian and set $\Sigma_k^{ndt} \leftarrow H^{-1} \max(\text{fitness}, 0.05)$
  Smooth and scale the covariance, enforce SPD, and publish `/ndt_correction`
  Store the current cloud as the next keyframe reference
end for

VI. Loop Closure and Visualization

Loop closure is attempted after the NDT factor has been inserted for the current node. Candidate previous nodes must be within 0.75 m of the current optimized pose and at least 10 indices behind the current node. The code ranks candidates by distance and tests at most 5 candidates. For each candidate, NDT is run between stored body-frame keyframe clouds. A loop is accepted only when the NDT fitness is below 0.12, the translation is below 1.25 m, and the yaw error relative to the graph-predicted relative motion is below 25°. The accepted loop covariance is scaled by 0.45 and inserted with a Huber robust kernel with $k = 1.345$, followed by 10 extra iSAM2 updates.

The graph publishes the optimized path, node-marker array, latest uncertainty ellipse, and per-keyframe evaluation snapshot. RViz markers include stored scan walls reprojected from optimized node poses, loop-closure edges, and covariance ellipses. Blue ellipses represent the full iSAM2 marginal covariance, green and red show odometry and NDT factor covariance, and the optional yellow ellipse shows a GTSAM local two-node marginal for the odom+NDT segment.

1) Algorithm 4: Loop Closure Detection and Insertion:

for each optimized keyframe node X_k after NDT fusion **do**
  if loop closure is disabled or k is smaller than the minimum index gap **then**
    Continue
  end if
  Find previous nodes within 0.75 m and at least 10 indices behind k
  Sort candidates by spatial distance and test at most five of them
  for each candidate i **do**
    if the stored scan cloud for i or k is missing **then**
      Continue
    end if
    Run NDT registration between the two stored body-frame clouds
    Compare NDT result with the graph-predicted relative motion
    if fitness < 0.12, translation < 1.25 m, and yaw error $< 25^\circ$ **then**
      Add robust Huber `BetweenFactor`$(X_i, X_k, z^{loop}, \Sigma^{loop})$
      Run 10 extra iSAM2 updates and publish updated markers
      Log the accepted loop edge and stop testing candidates
    end if
  end for
end for

TABLE I
REPRESENTATIVE MARGINAL COVARIANCE STATISTICS FROM THE REAL ROSBAG.

KF	Event	σ_x [m]	σ_y [m]	σ_θ [°]	σ_{pos} [m]
2	Early NDT	0.062	0.065	3.5	0.090
12	Mid run	0.163	0.281	8.3	0.325
16	Peak uncertainty	0.320	0.223	9.7	0.390
21	LC X11→X21	0.186	0.236	9.4	0.300
24	LC X2→X24	0.119	0.126	6.6	0.173
26	End of run	0.108	0.105	5.6	**0.151**

VII. EVALUATION METHOD

The current evaluator subscribes to `/slam_kf_eval_snapshot`. The reported real-data evaluation uses the bag `rosbag2_2026_03_25-06_01_48`, which lasts 150.37 s and contains 44,905 messages. In addition to `/cmd_vel`, the bag records `/odom`, `/scan`, `/imu`, wheel state topics, camera topics, TF, and the SLAM outputs generated during the run. The relevant Graph SLAM topics include 26 `/step_covariance` messages, 25 `/ndt_correction` messages, 25 `/slam_kf_eval_snapshot` messages, 51 `/slam_graph_path` publications, and 51 `/slam_node_markers` publications. The analysis therefore uses the recorded graph trajectory, covariance snapshots, NDT corrections, and loop closure markers directly from the rosbag.

The active covariance metric is the iSAM2 marginal standard deviation after each post-NDT keyframe update. The positional marginal standard deviation is summarized as

$$\sigma_{pos} = \sqrt{\Sigma_{xx} + \Sigma_{yy}}. \tag{15}$$

The recorded `/rosout` messages are used to identify accepted loop closures, while `/slam_node_markers` provides the loop-closure edge markers and the stored scan-history points used in the trajectory figures.

VIII. RESULTS

The real rosbag produced a compact graph with 27 optimized poses: the anchored prior node plus 26 covariance-triggered keyframe steps. The first NDT trigger only initializes the reference scan cloud, so 25 NDT correction factors are recorded. Six loop closures were accepted near the end of the run: `X11→X21`, `X10→X22`, `X4→X23`, `X2→X24`, `X1→X25`, and `X1→X26`. Their NDT fitness values were between 0.001 and 0.097, all below the configured loop-closure gate of 0.12.

Figure 4 shows the optimized graph trajectory, the aligned raw wheel-odometry trajectory, the stored scan-history wall points, and the recorded loop-closure marker edges. This figure comes directly from the bag topics `/odom`, `/slam_graph_path`, and `/slam_node_markers`.

The first accepted loop closure, `X11→X21`, reduced the keyframe marginal position standard deviation from 0.360 m at KF20 to 0.300 m at KF21:

$$\frac{0.360 - 0.300}{0.360} \times 100 \approx 16.5\%. \tag{16}$$

The later loop closures continued to constrain the graph, reducing σ_{pos} to 0.173 m at KF24 and to approximately 0.151 m at KF25–26. The determinant of the marginal covariance also dropped from 1.01×10^{-4} at KF20 to 1.19×10^{-6} at the final keyframe, which is about an 85-fold reduction.

IX. DISCUSSION

The results match the expected behaviour of a pose-graph SLAM system. Before loop closure, the graph is mostly a chain of relative constraints. Even though each edge has both odometry and NDT information, the absolute location of the latest node remains weakly constrained relative to the start, so the marginal uncertainty grows with path length. In the real rosbag, this is visible from KF2 to KF16, where σ_{pos} rises from 0.090 m to 0.390 m. When the robot revisits previously mapped areas, loop factors connect the recent nodes back to earlier nodes and iSAM2 redistributes the accumulated error. The accepted loop edges from KF21 to KF26 explain the steady covariance reduction at the end of the experiment.

NDT contributes in two ways. First, it provides an additional local relative measurement for every accepted keyframe. Second, it validates loop closures using stored scan clouds, reducing the probability of accepting a spatially close but geometrically inconsistent place. Figure 7 compares the local NDT correction covariance with the final GTSAM marginal covariance at each keyframe. These quantities are not expected to be identical: the NDT covariance belongs to one scan-matching factor, whereas the GTSAM marginal covariance is the posterior uncertainty after all graph constraints are fused. The decrease in the GTSAM marginal determinant after KF21 shows the effect of the accepted loop-closure constraints on the optimized graph.

The covariance-triggered keyframe policy is useful because keyframe creation depends on uncertainty rather than only distance or time. Turns and uncertain motion add covariance faster, while stationary jitter is filtered out by motion gates. This keeps the graph informative without creating unnecessary nodes when the robot is not moving.

X. LIMITATIONS

The reported real-data evaluation uses recorded SLAM outputs from one rosbag, so it verifies the behaviour of the implemented system on that run but does not replace a broad benchmark over many environments. The bag does not provide an independent motion-capture or survey-grade ground truth trajectory; therefore the quantitative discussion focuses on graph marginal covariance, NDT correction consistency, and loop-closure behaviour rather than absolute trajectory error. NDT can still be weak in feature-poor or repetitive environments, and spatial loop closure can be vulnerable to perceptual aliasing if two places have similar scan geometry.

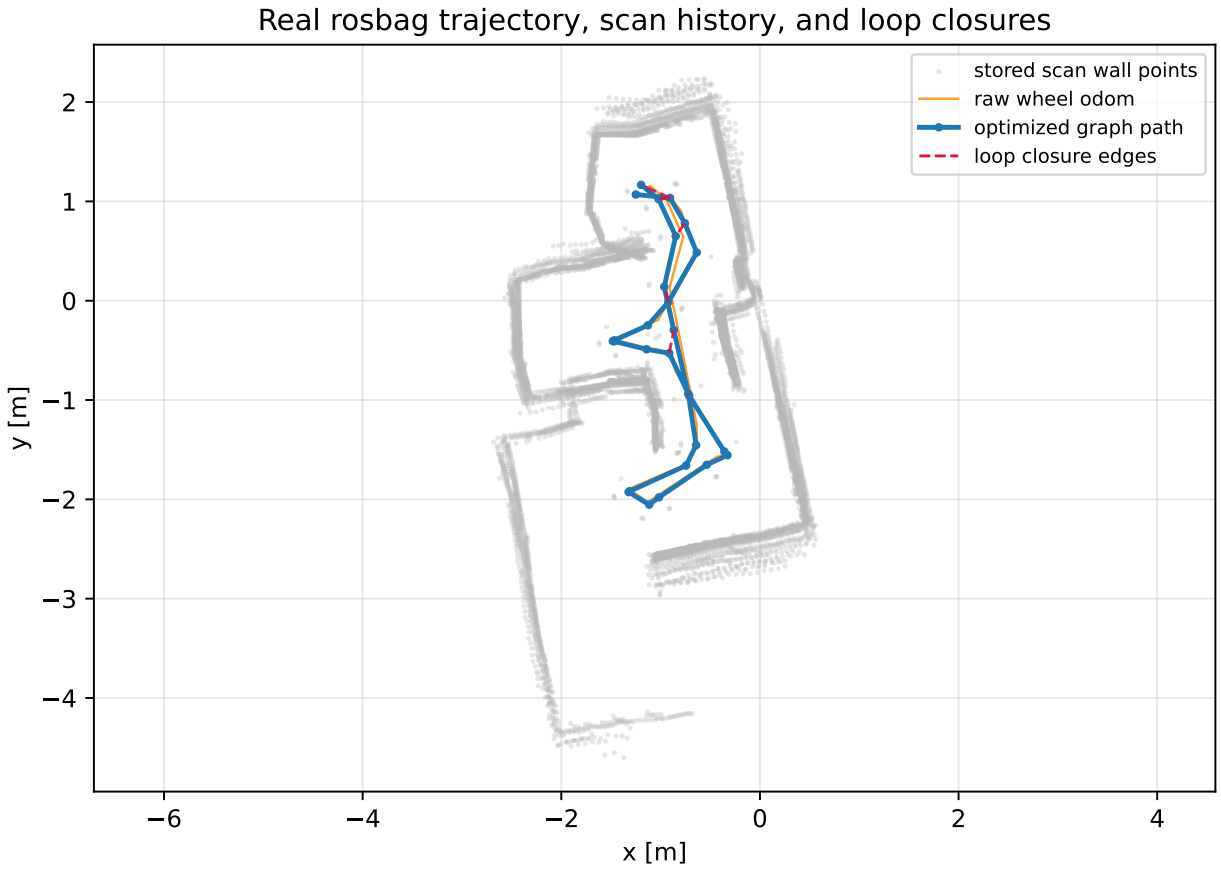


Fig. 4. Real rosbag trajectory and map-style visualization. The optimized graph path is plotted with the raw wheel-odometry trajectory, stored scan wall points, and the loop-closure edges recorded in `/slam_node_markers`.

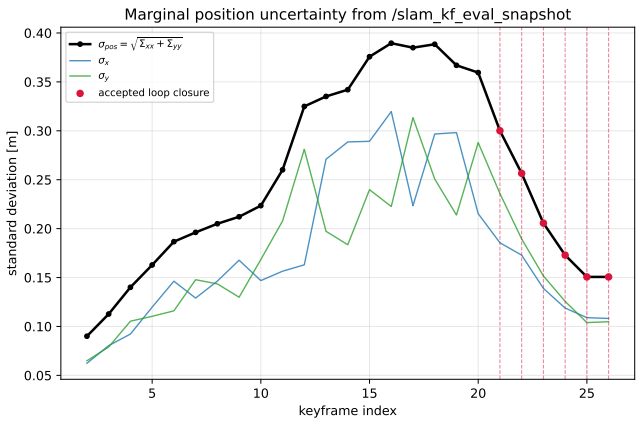


Fig. 5. Marginal position uncertainty from the real rosbag. The uncertainty increases during open-loop motion, reaches its maximum near KF16, and decreases after the sequence of accepted loop closures from KF21 to KF26.

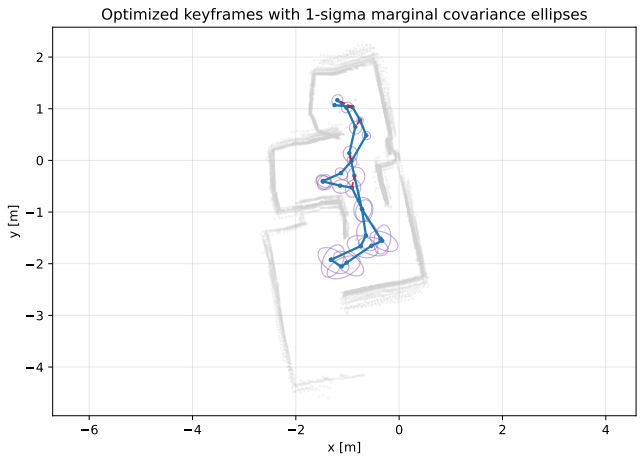


Fig. 6. Optimized keyframes with 1-σ marginal covariance ellipses from `/slam_kf_eval_snapshot`. The ellipses become tighter after the loop closures connect the final part of the path back to earlier nodes.

A complete benchmarking study should add absolute trajectory error, relative pose error, and consistency tests such as NEES when ground truth is available.

XI. Conclusion

This paper documented the current Graph SLAM implementation in the `graph_slam_project` package. The system

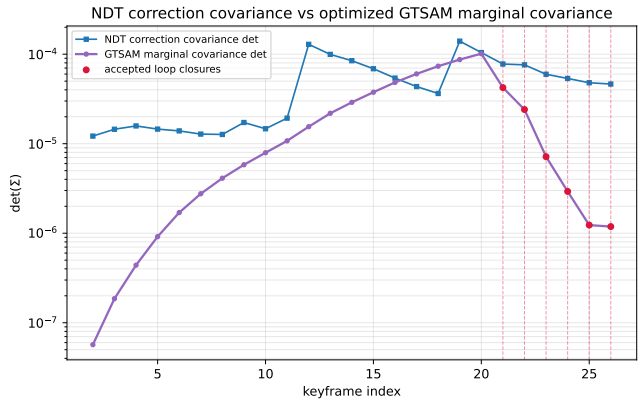


Fig. 7. NDT correction covariance compared with the optimized GTSAM marginal covariance. The NDT covariance describes a local scan-matching factor, while the GTSAM marginal covariance describes the posterior uncertainty of the optimized graph node after odometry, NDT, and loop-closure constraints are fused.

creates keyframes using an SE(2) covariance integrator, adds odometry and NDT factors to an iSAM2 graph, validates loop closures using NDT over stored keyframe clouds, and visualizes full-graph marginal covariance. In the real TurtleBot 4 rosbag, the system processed 26 keyframe steps, fused 25 NDT corrections, and accepted six loop closures. The marginal position uncertainty peaked at approximately 0.390 m and decreased to 0.151 m by the end of the run, confirming that the implemented loop-closure factors significantly tightened the graph uncertainty.

ACKNOWLEDGMENT

The authors thank Prof. Pedro Ridao Rodriguez for his guidance and supervision throughout this project.

REFERENCES

[1] F. Dellaert and M. Kaess, "Factor graphs for robot perception," *Foundations and Trends in Robotics*, vol. 6, no. 1–2, pp. 1–139, 2017.

[2] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. Leonard, and F. Dellaert, "iSAM2: Incremental smoothing and mapping using the Bayes tree," *The International Journal of Robotics Research*, vol. 31, no. 2, pp. 216–235, 2012.

[3] M. Magnusson, *The Three-Dimensional Normal-Distributions Transform: An Efficient Representation for Registration, Surface Analysis, and Loop Detection*. Ph.D. dissertation, Orebro University, 2009.

[4] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005.